LendingClub Dataset

Deep Data Profile, Quality Audit & Exploratory Analysis

Generated: 2026-10-29 14:30

Source file: accepted_2007_to_2018Q4.csv

Dataset size: 314,168 rows × 162 columns

Purpose. Document dataset structure, quality risks, distributions, relationships, and modeling considerations before feature engineering or supervised learning.

Critical warning. Some variables are measured after loan origination or after repayment outcomes occur. A default prediction model must use only information available at its defined prediction time.

1. Executive summary

Metric	Result
Rows	314,168
Columns	162
Duplicate full rows	0
Numeric columns	115
Categorical/text columns	36
Total memory (MB)	830.97
Rows with any missing data	314,168 (100.00%)
Columns with any missing data	142

This summary is computed from the loaded file. It is not a statement about any other version of the LendingClub dataset.

2. Schema inventory

The schema inventory records inferred type, semantic hint, missingness, unique values, and example values. Semantic hints are heuristics based on names and observed values.

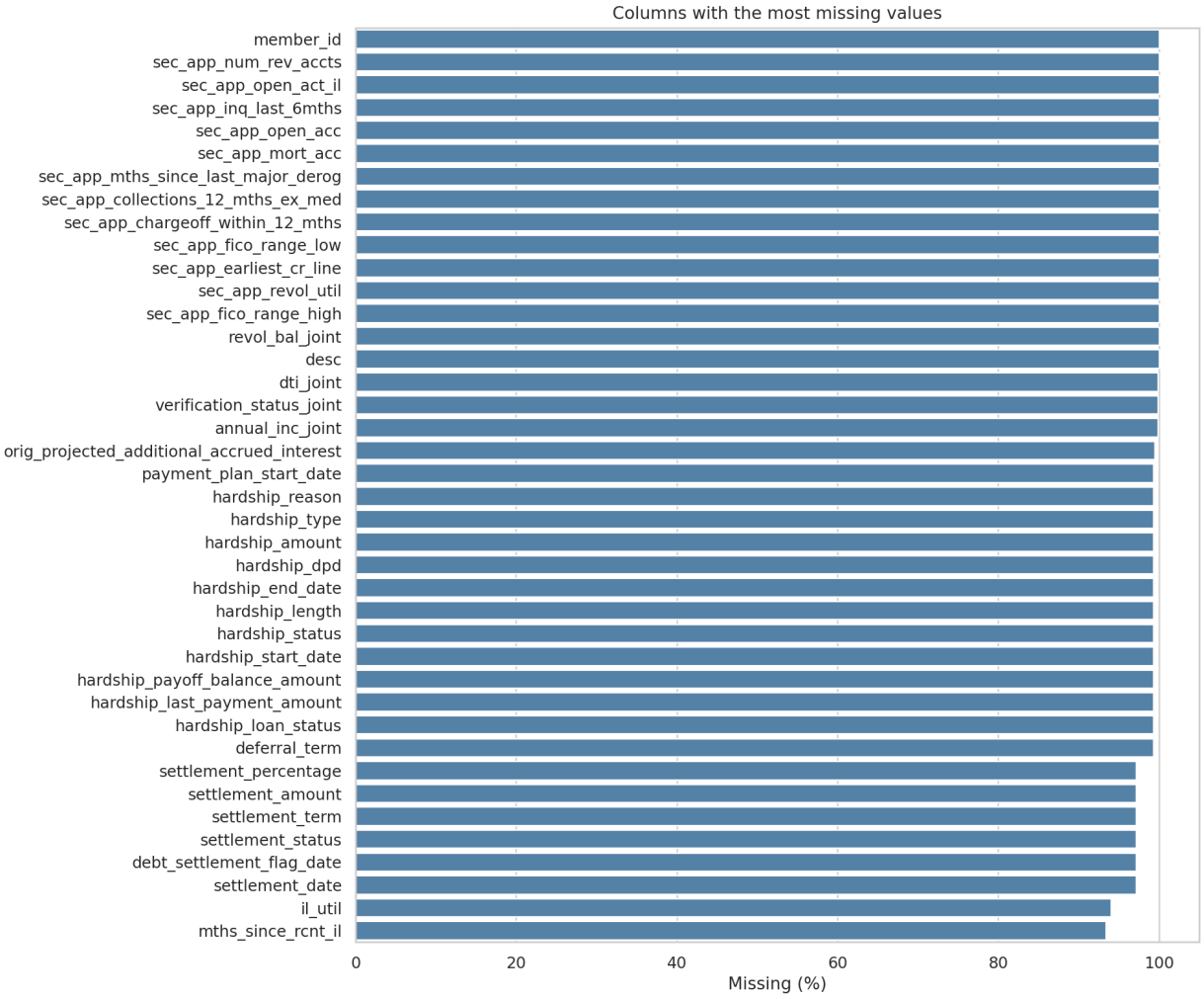
column	dtype_inferred	semantic_hint	missing_pct	n_unique_non_null	sample_values
member_id	float64	identifier / link	100.0	0	
revol_bal_joint	float64	numeric binary / low-cardinality	100.0	0	
sec_app_chargeoff_within_12_mths	float64	numeric binary / low-cardinality	100.0	0	
sec_app_collections_12_mths_ex_med	float64	numeric binary / low-cardinality	100.0	0	
sec_app_earliest_cr_line	float64	date-like (inspect format)	100.0	0	
sec_app_fico_range_high	float64	numeric binary / low-cardinality	100.0	0	
sec_app_fico_range_low	float64	numeric binary / low-cardinality	100.0	0	
sec_app_inq_last_6mths	float64	numeric binary / low-cardinality	100.0	0	
sec_app_mort_acc	float64	numeric binary / low-cardinality	100.0	0	
sec_app_mths_since_last_major_derog	float64	date-like (inspect format)	100.0	0	
sec_app_num_rev_accts	float64	numeric binary / low-cardinality	100.0	0	
sec_app_open_acc	float64	numeric binary / low-cardinality	100.0	0	
sec_app_open_act_il	float64	numeric binary / low-cardinality	100.0	0	
sec_app_revol_util	float64	numeric binary / low-cardinality	100.0	0	
desc	object	categorical / text	99.993	19	We knew that using our credit cards to finance an
dti_joint	float64	numeric measure / count / code (inspect)	99.838	449	13.85 | 16.4 | 15.4 | 25.44
annual_inc_joint	float64	numeric measure / count / code (inspect)	99.837	308	71000.0 | 100000.0 | 77000.0 | 113314.0
verification_status_joint	object	categorical / status	99.837	1	Not Verified | Not Verified | Not Verified | Not Verif
orig_projected_additional_accrued_interest	float64	numeric measure / count / code (inspect)	99.456	1641	442.41 | 641.94 | 108.21 | 503.16
deferral_term	float64	categorical / status	99.329	1	3.0 | 3.0 | 3.0 | 3.0
hardship_amount	float64	numeric measure / count / code (inspect)	99.329	2010	47.01 | 147.47 | 213.98 | 36.07
hardship_dpd	float64	date-like (inspect format)	99.329	31	26.0 | 0.0 | 18.0 | 16.0
hardship_end_date	object	date-like (inspect format)	99.329	27	Dec-2017 | Nov-2017 | Dec-2017 | Oct-2017
hardship_last_payment_amount	float64	numeric measure / count / code (inspect)	99.329	1984	61.23 | 6.27 | 275.35 | 55.51
hardship_length	float64	numeric binary / low-cardinality	99.329	1	3.0 | 3.0 | 3.0 | 3.0
hardship_loan_status	object	categorical / status	99.329	4	Late (16-30 days) | Current | Late (16-30 days) | L
hardship_payoff_balance_amount	float64	numeric measure / count / code (inspect)	99.329	2106	3058.14 | 12014.81 | 18761.48 | 4801.21
hardship_reason	object	categorical / status	99.329	9	NATURAL_DISASTER | NATURAL_DISASTER |
hardship_start_date	object	date-like (inspect format)	99.329	26	Sep-2017 | Sep-2017 | Sep-2017 | Jul-2017
hardship_status	object	categorical / status	99.329	3	BROKEN | COMPLETED | COMPLETED | COMP
hardship_type	object	categorical / status	99.329	1	INTEREST ONLY-3 MONTHS DEFERRAL | INTE
payment_plan_start_date	object	date-like (inspect format)	99.329	27	Oct-2017 | Sep-2017 | Sep-2017 | Aug-2017
debt_settlement_flag_date	object	date-like (inspect format)	97.166	41	Nov-2017 | May-2018 | Feb-2019 | May-2018
settlement_amount	float64	numeric measure / count / code (inspect)	97.166	7480	9915.0 | 2761.0 | 4382.0 | 3513.0
settlement_date	object	date-like (inspect format)	97.166	44	Sep-2017 | Nov-2017 | Jan-2018 | May-2018
settlement_percentage	float64	numeric measure / count / code (inspect)	97.166	677	55.0 | 50.01 | 44.99 | 45.01
settlement_status	object	categorical / status	97.166	3	COMPLETE | BROKEN | COMPLETE | ACTIVE
settlement_term	float64	categorical / status	97.166	33	1.0 | 10.0 | 12.0 | 16.0
il_util	float64	numeric measure / count / code (inspect)	94.074	160	36.0 | 73.0 | 73.0 | 70.0
mths_since_rcnt_il	float64	numeric measure / count / code (inspect)	93.376	201	21.0 | 19.0 | 19.0 | 23.0
all_util	float64	numeric measure / count / code (inspect)	93.197	139	34.0 | 29.0 | 65.0 | 45.0
inq_fi	float64	numeric measure / count / code (inspect)	93.197	18	3.0 | 0.0 | 2.0 | 0.0
inq_last_12m	float64	numeric measure / count / code (inspect)	93.197	28	4.0 | 6.0 | 1.0 | 0.0
max_bal_bc	float64	numeric measure / count / code (inspect)	93.197	10707	722.0 | 6472.0 | 2081.0 | 6987.0
open_acc_6m	float64	numeric measure / count / code (inspect)	93.197	13	2.0 | 1.0 | 0.0 | 1.0
open_act_il	float64	numeric measure / count / code (inspect)	93.197	35	2.0 | 1.0 | 1.0 | 1.0
open_il_12m	float64	numeric measure / count / code (inspect)	93.197	12	0.0 | 0.0 | 0.0 | 0.0
open_il_24m	float64	numeric measure / count / code (inspect)	93.197	17	1.0 | 1.0 | 4.0 | 1.0
open_rv_12m	float64	numeric measure / count / code (inspect)	93.197	18	3.0 | 2.0 | 0.0 | 1.0
open_rv_24m	float64	numeric measure / count / code (inspect)	93.197	28	3.0 | 3.0 | 2.0 | 1.0
total_bal_il	float64	numeric measure / count / code (inspect)	93.197	17030	4981.0 | 18005.0 | 10827.0 | 12609.0

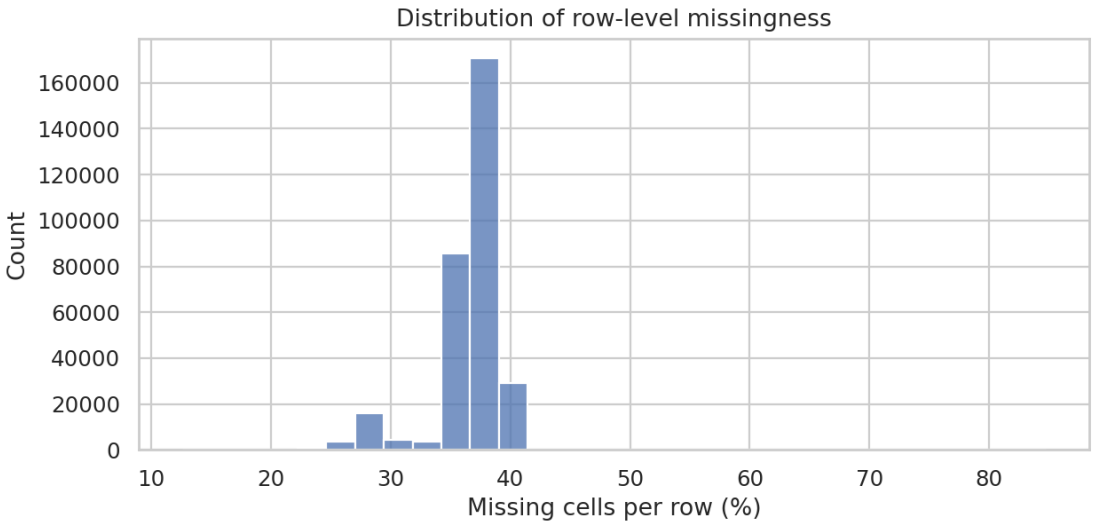
column	dtype_inferred	semantic_hint	missing_pct	n_unique_non_null	sample_values
total_cu_tl	float64	numeric measure / count / code (inspect)	93.197	33	1.0 | 0.0 | 5.0 | 1.0
next_pymnt_d	object	date-like (inspect format)	88.627	6	Apr-2019 | Apr-2019 | Apr-2019 | Apr-2019
mths_since_last_record	float64	numeric measure / count / code (inspect)	82.271	121	106.0 | 75.0 | 30.0 | 2.0
mths_since_recent_bc_dlq	float64	date-like (inspect format)	74.39	146	69.0 | 12.0 | 48.0 | 30.0
mths_since_last_major_derog	float64	date-like (inspect format)	70.786	148	30.0 | 3.0 | 75.0 | 54.0
mths_since_recent_revol_delinq	float64	date-like (inspect format)	64.043	140	69.0 | 6.0 | 12.0 | 42.0
mths_since_last_delinq	float64	date-like (inspect format)	48.485	134	30.0 | 6.0 | 12.0 | 49.0
mths_since_recent_inq	float64	numeric measure / count / code (inspect)	10.725	26	4.0 | 0.0 | 10.0 | 1.0
emp_title	object	categorical / status	5.715	95534	leadman | Engineer | truck driver | Information Sys

3. Missingness

column	missing_count	missing_pct	non_missing_count
member_id	314168	100.0	0
sec_app_num_rev_accts	314168	100.0	0
sec_app_open_act_il	314168	100.0	0
sec_app_inq_last_6mths	314168	100.0	0
sec_app_open_acc	314168	100.0	0
sec_app_mort_acc	314168	100.0	0
sec_app_mths_since_last_major_derog	314168	100.0	0
sec_app_collections_12_mths_ex_med	314168	100.0	0
sec_app_chargeoff_within_12_mths	314168	100.0	0
sec_app_fico_range_low	314168	100.0	0
sec_app_earliest_cr_line	314168	100.0	0
sec_app_revol_util	314168	100.0	0
sec_app_fico_range_high	314168	100.0	0
revol_bal_joint	314168	100.0	0
desc	314145	99.99267907616307	23
dti_joint	313659	99.83798477247842	509
verification_status_joint	313657	99.83734817040563	511
annual_inc_joint	313657	99.83734817040563	511
orig_projected_additional_accrued_interest	312459	99.45602352881261	1709
payment_plan_start_date	312061	99.32933971633011	2107
hardship_reason	312061	99.32933971633011	2107
hardship_type	312061	99.32933971633011	2107
hardship_amount	312061	99.32933971633011	2107
hardship_dpd	312061	99.32933971633011	2107
hardship_end_date	312061	99.32933971633011	2107
hardship_length	312061	99.32933971633011	2107
hardship_status	312061	99.32933971633011	2107
hardship_start_date	312061	99.32933971633011	2107
hardship_payoff_balance_amount	312061	99.32933971633011	2107
hardship_last_payment_amount	312061	99.32933971633011	2107
hardship_loan_status	312061	99.32933971633011	2107
deferral_term	312061	99.32933971633011	2107
settlement_percentage	305263	97.1655292709633	8905
settlement_amount	305263	97.1655292709633	8905
settlement_term	305263	97.1655292709633	8905
settlement_status	305263	97.1655292709633	8905
debt_settlement_flag_date	305263	97.1655292709633	8905
settlement_date	305263	97.1655292709633	8905

column	missing_count	missing_pct	non_missing_count
il_util	295551	94.07418960556136	18617
mths_since_rcnt_il	293358	93.37615543276209	20810
total_cu_tl	292796	93.19727025031193	21372
open_acc_6m	292796	93.19727025031193	21372
inq_fi	292796	93.19727025031193	21372
inq_last_12m	292796	93.19727025031193	21372
total_bal_il	292796	93.19727025031193	21372




Distribution of row-level missingness
Count
160000
140000
120000
100000
80000
60000
40000
20000
0
10
20
30
40
50
60
70
80
Missing cells per row (%)

4. Numeric profile

Descriptive statistics include quantiles, skewness, kurtosis, zero share, missingness, and distinct values. Extreme values are flagged for investigation, not automatic removal.

column	count	mean	std	min	1%	5%	25%	50%	75%
id	314168.0	58224028.400107585	6640320.8102135435	56705.0	46367127.71	46803963.7	53402416.25	59121330.5	63505072.5
member_id	0.0								
loan_amnt	314168.0	15216.877196277152	8608.328191370274	1000.0	1925.0	3775.0	8450.0	14000.0	20000.0
funded_amnt	314168.0	15216.877196277152	8608.328191370274	1000.0	1925.0	3775.0	8450.0	14000.0	20000.0
funded_amnt_inv	314168.0	15209.839621196237	8604.340646941044	900.0	1925.0	3775.0	8450.0	14000.0	20000.0
int_rate	314168.0	12.509651874156505	4.314629341241656	5.32	5.32	6.24	9.17	12.29	15.41
installment	314168.0	440.6652652402536	245.98594595368897	14.77	62.58	125.68	261.88	382.55	578.68
annual_inc	314168.0	77435.89938978506	77296.43766727863	0.0	19000.0	28800.0	47000.0	65000.0	92000.0
dti	314166.0	19.1650938994035	8.97537426176314	0.0	2.35	5.64	12.58	18.62	25.37
delinq_2yrs	314167.0	0.348980001082227	0.9283014438907669	0.0	0.0	0.0	0.0	0.0	0.0
fico_range_low	314167.0	694.1220911171447	30.47534724589945	660.0	660.0	660.0	670.0	685.0	710.0
fico_range_high	314167.0	698.122227987026	30.47602710368508	664.0	664.0	664.0	674.0	689.0	714.0
inq_last_6mths	314167.0	0.5702667689477252	0.8666501440275945	0.0	0.0	0.0	0.0	0.0	1.0
mths_since_last_delinq	161845.0	34.09198924897278	22.027769958996966	0.0	1.0	5.0	15.0	31.0	50.0
mths_since_last_record	55698.0	66.1779417573342	24.967975761380586	0.0	9.0	23.0	50.0	67.0	82.0
open_acc	314167.0	11.973310373145493	5.659049200031275	1.0	3.0	5.0	8.0	11.0	15.0
pub_rec	314167.0	0.234601342598045	0.6512763850438645	0.0	0.0	0.0	0.0	0.0	0.0
revol_bal	314167.0	17742.96247218836	24710.99099017361	0.0	382.65999999999985	2052.0	6432.0	11933.0	21385.5
revol_util	314044.0	53.275488785011035	23.952172999505116	0.0	2.5	13.1	35.4	53.5	71.6
total_acc	314167.0	25.41318470749633	12.080232470183015	4.0	6.0	9.0	17.0	24.0	32.0
out_prncp	314167.0	843.488884414977	2573.762670626898	0.0	0.0	0.0	0.0	0.0	0.0
out_prncp_inv	314167.0	843.0307338453754	2572.2845814232633	0.0	0.0	0.0	0.0	0.0	0.0
total_pymnt	314167.0	15620.600572599607	9946.86477775298	0.0	1306.0757999999998	3116.07462486587	7932.22866081165	13389.62	21605.545
total_pymnt_inv	314167.0	15613.525950497666	9942.257051421066	0.0	1304.4928	3114.3780000000006	7926.985000000001	13385.45	21598.785
total_rec_prncp	314167.0	12428.012034459378	8380.264877323318	0.0	607.9398	1800.86	6000.0	10200.0	17125.0
total_rec_int	314167.0	2959.6135208981223	2945.0064046134476	0.0	78.7766	310.83	976.23	1937.04	3956.11
total_rec_late_fee	314167.0	1.9281877605537365	13.38024221008283	-9.5e-09	0.0	0.0	0.0	0.0	0.0
recoveries	314167.0	231.0468259237922	970.5444433528655	0.0	0.0	0.0	0.0	0.0	0.0
collection_recovery_fee	314167.0	41.236442168328516	173.98212970106349	0.0	0.0	0.0	0.0	0.0	0.0
last_pymnt_amnt	314167.0	4252.03519436872	6378.973535636444	0.0	1.26	79.93	340.23	782.15	6088.33
last_fico_range_high	314167.0	677.3599327746072	79.90866395678096	0.0	499.0	524.0	629.0	694.0	734.0
last_fico_range_low	314167.0	661.4501523075307	128.01142770265025	0.0	0.0	520.0	625.0	690.0	730.0
collections_12_mths_ex_med	314167.0	0.0212753089917146	0.160943634168192	0.0	0.0	0.0	0.0	0.0	0.0
mths_since_last_major_derog	91781.0	43.947026072934484	21.403714279824538	0.0	3.0	9.0	27.0	44.0	62.0
policy_code	314167.0	1.0	0.0	1.0	1.0	1.0	1.0	1.0	1.0
annual_inc_joint	511.0	109981.0115851272	52730.37984721528	17950.0	29779.2	47488.0	76032.5	101771.0	132800.0
dti_joint	509.0	18.310117878192536	7.169233422067742	3.0	4.0256	7.492	13.2	17.61	22.65
acc_now_delinq	314167.0	0.005799463342744464	0.08178585942783105	0.0	0.0	0.0	0.0	0.0	0.0
tot_coll_amt	314167.0	258.6984565533617	2166.4839280324395	0.0	0.0	0.0	0.0	0.0	0.0
tot_cur_bal	314167.0	140819.279236839	155775.19922130118	0.0	3742.66	9647.0	31316.0	80709.0	208453.0
open_acc_6m	21372.0	1.1090211491671347	1.2426750562102509	0.0	0.0	0.0	0.0	1.0	2.0
open_act_il	21372.0	2.928832116788321	3.0899871930167144	0.0	0.0	0.0	1.0	2.0	4.0
open_il_12m	21372.0	0.7616507580011229	0.9960346479643885	0.0	0.0	0.0	0.0	0.0	1.0
open_il_24m	21372.0	1.674574209245742	1.6887251954652032	0.0	0.0	0.0	0.0	1.0	2.0
mths_since_rcnt_il	20810.0	20.91268620855358	27.20908126109245	0.0	1.0	2.0	6.0	12.0	23.0
total_bal_il	21372.0	36552.811388732924	43103.83361919471	0.0	0.0	0.0	10251.5	24684.5	47858.0
il_util	18617.0	71.58049094913251	23.021964104681754	0.0	5.0	28.0	59.0	75.0	88.0
open_rv_12m	21372.0	1.389060452929066	1.5201293497880226	0.0	0.0	0.0	0.0	1.0	2.0
open_rv_24m	21372.0	2.9754819389855887	2.6318859876068483	0.0	0.0	0.0	1.0	2.0	4.0
max_bal_bc	21372.0	5887.979739846528	5284.701239120277	0.0	0.0	516.55	2411.0	4483.0	7771.5
all_util	21372.0	60.88199513381995	20.01384153443746	0.0	11.0	26.0	48.0	62.0	75.0
total_rev_hi_lim	314167.0	34014.35264365767	39102.054036656875	0.0	3000.0	6200.0	14500.0	24800.0	42000.0

column	count	mean	std	min	1%	5%	25%	50%	75%
inq_fi	21372.0	0.9439453490548381	1.4468715236470169	0.0	0.0	0.0	0.0	0.0	1.0
total_cu_tl	21372.0	1.5370578326782707	2.7233391655444144	0.0	0.0	0.0	0.0	0.0	2.0
inq_last_12m	21372.0	2.234091334456298	2.4802418996648754	0.0	0.0	0.0	0.0	2.0	3.0

5. Outlier screening

column	iqr_outlier_count	iqr_outlier_pct	robust_z_gt_3_5_count	iqr_lower_fence	iqr_upper_fence
inq_fi	2422	11.333	0	-1.5	2.5
mths_since_rcnt_il	2048	9.841	2048	-19.5	48.5
mths_since_recent_bc	29660	9.526	31443	-28.5	63.5
bc_open_to_buy	27332	8.781	33292	-14172.0	27372.0
last_pymnt_amnt	26559	8.454	98211	-8281.920000000002	14710.480000000001
mo_sin_rcnt_rev_tl_op	26001	8.276	27374	-14.0	34.0
settlement_percentage	736	8.265	4165	37.5	57.5
total_cu_tl	1651	7.725	0	-3.0	5.0
mo_sin_rcnt_tl	20325	6.469	20325	-7.5	20.5
mo_sin_old_il_acct	19419	6.362	4862	23.0	231.0
num_il_tl	19807	6.305	8889	-6.5	21.5
pct_tl_nvr_dlq	19622	6.246	39254	77.25000000000001	113.64999999999999
revol_bal	19283	6.138	16955	-15998.25	43815.75
total_bc_limit	19064	6.068	17465	-23200.0	60000.0
total_rec_int	19054	6.065	20795	-3493.5899999999997	8425.93
total_rev_hi_lim	18843	5.998	16146	-26750.0	83250.0
total_bal_il	1273	5.956	1043	-46158.25	104267.75
total_bal_ex_mort	17983	5.724	14280	-42449.0	131351.0
open_act_il	1218	5.699	1514	-3.5	8.5
avg_cur_bal	17240	5.488	25571	-19033.5	40314.5
open_il_12m	1126	5.269	0	-1.5	2.5
max_bal_bc	1120	5.241	900	-5629.75	15812.25
num_bc_sats	15708	5.0	4091	-1.5	10.5
total_il_high_credit_limit	15672	4.988	11200	-48808.75	123481.25
annual_inc	15577	4.958	12105	-20500.0	159500.0
num_bc_tl	14351	4.568	3994	-2.5	17.5
inq_last_6mths	12707	4.045	0	-1.5	2.5
inq_last_12m	823	3.851	823	-4.5	7.5
open_rv_24m	820	3.837	1247	-3.5	8.5
tot_cur_bal	11440	3.641	19348	-234389.5	474158.5
tot_hi_cred_lim	11042	3.515	14496	-247704.75	550441.25
num_actv_rev_tl	10942	3.483	5878	-3.0	13.0
hardship_last_payment_amount	70	3.322	57	-298.6225000000001	614.9175000000001
open_il_24m	684	3.2	375	-3.0	5.0
total_rec_prncp	9843	3.133	0	-10687.5	33812.5
num_rev_tl_bal_gt_0	9801	3.12	5040	-3.0	13.0
hardship_amount	64	3.037	37	-145.90750000000003	358.2325
orig_projected_additional_accrued_interest	50	2.926	25	-433.93500000000006	1065.4650000000001
num_tl_op_past_12m	8961	2.852	4874	-2.0	6.0
fico_range_low	8900	2.833	4754	610.0	770.0

6. Categorical and text profile

column	missing_pct	n_unique	top_value	top_value_count	top_value_pct_nonmissing	high_cardinality_candi
url	0.0	314168	https://lendingclub.com/browse/loanDetail.action?loan_id=45394672	1	0.0	True
emp_title	5.715	95534	Teacher	6186	2.088	True
zip_code	0.0	904	750xx	3298	1.05	False
earliest_cr_line	0.0	658	Aug-2002	2395	0.762	False
addr_state	0.0	49	CA	42895	13.654	False
last_pymnt_d	0.076	48	Mar-2019	32764	10.437	False
last_credit_pull_d	0.001	48	Mar-2019	133382	42.456	False
settlement_date	97.166	44	Mar-2018	500	5.615	False
debt_settlement_flag_date	97.166	41	Oct-2018	653	7.333	False
sub_grade	0.0	35	C1	19805	6.304	False
hardship_end_date	99.329	27	Dec-2017	429	20.361	False
payment_plan_start_date	99.329	27	Oct-2017	386	18.32	False
hardship_start_date	99.329	26	Sep-2017	551	26.151	False
desc	99.993	19		5	21.739	False
title	0.042	14	Debt consolidation	184024	58.6	False
purpose	0.0	13	debt_consolidation	184075	58.591	False
emp_length	5.704	11	10+ years	105677	35.672	False
hardship_reason	99.329	9	NATURAL_DISASTER	580	27.527	False
issue_d	0.0	9	Oct-2015	48631	15.479	False
grade	0.0	7	C	90025	28.655	False
loan_status	0.0	7	Fully Paid	222280	70.752	False
next_pymnt_d	88.627	6	Apr-2019	35708	99.941	False
hardship_loan_status	99.329	4	Late (16-30 days)	937	44.471	False
home_ownership	0.0	4	MORTGAGE	155451	49.48	False
hardship_status	99.329	3	COMPLETED	1666	79.07	False
settlement_status	97.166	3	COMPLETE	4473	50.23	False
verification_status	0.0	3	Source Verified	136250	43.369	False
term	0.0	2	36 months	211558	67.339	False
pymnt_plan	0.0	2	n	314138	99.99	False
initial_list_status	0.0	2	w	214287	68.208	False
application_type	0.0	2	Individual	313656	99.837	False
hardship_flag	0.0	2	N	314124	99.986	False
debt_settlement_flag	0.0	2	N	305262	97.166	False
verification_status_joint	99.837	1	Not Verified	511	100.0	False
hardship_type	99.329	1	INTEREST ONLY-3 MONTHS DEFERRAL	2107	100.0	False
disbursement_method	0.0	1	Cash	314167	100.0	False

7. Text hygiene

column	values_with_outer_whitespace	blank_strings	case_variant_count_approx	n_unique_original	n_unique_after_strip
term	314168	0	0	2	2
emp_title	20837	0	-13893	95534	90419
desc	12	5	0	19	19
grade	0	0	0	7	7
sub_grade	0	0	0	35	35
emp_length	0	0	0	11	11
verification_status	0	0	0	3	3
home_ownership	0	0	0	4	4
issue_d	0	0	0	9	9
loan_status	0	0	0	7	7
pymnt_plan	0	0	0	2	2
url	0	0	0	314168	314168

column	values_with_outer_whitespace	blank_strings	case_variant_count_approx	n_unique_original	n_unique_after_strip
purpose	0	0	0	13	13
title	0	0	0	14	14
zip_code	0	0	0	904	904
addr_state	0	0	0	49	49
earliest_cr_line	0	0	0	658	658
initial_list_status	0	0	0	2	2
last_pymnt_d	0	0	0	48	48
next_pymnt_d	0	0	0	6	6
last_credit_pull_d	0	0	0	48	48
application_type	0	0	0	2	2
verification_status_joint	0	0	0	1	1
hardship_flag	0	0	0	2	2
hardship_type	0	0	0	1	1
hardship_reason	0	0	0	9	9
hardship_status	0	0	0	3	3
hardship_start_date	0	0	0	26	26
hardship_end_date	0	0	0	27	27
payment_plan_start_date	0	0	0	27	27
hardship_loan_status	0	0	0	4	4
disbursement_method	0	0	0	1	1
debt_settlement_flag	0	0	0	2	2
debt_settlement_flag_date	0	0	0	41	41
settlement_status	0	0	0	3	3

8. Date parsing and coverage

column	original_non_null	parsed_non_null	parse_success_pct_of_nonnull	min_date	max_date
issue_d	314168	314168	100.0	2015-04-01	2015-12-01
earliest_cr_line	314167	314167	100.0	1950-01-01	2012-11-01
last_pymnt_d	313928	313928	100.0	2015-04-01	2019-03-01
next_pymnt_d	35729	35729	100.0	2018-08-01	2019-05-01
last_credit_pull_d	314164	314164	100.0	2015-04-01	2019-03-01
sec_app_earliest_cr_line	0	0	0.0	NaT	NaT
hardship_start_date	2107	2107	100.0	2017-02-01	2019-03-01
hardship_end_date	2107	2107	100.0	2017-03-01	2019-06-01
payment_plan_start_date	2107	2107	100.0	2017-02-01	2019-04-01
debt_settlement_flag_date	8905	8905	100.0	2015-09-01	2019-03-01
settlement_date	8905	8905	100.0	2015-08-01	2019-03-01

9. Identifiers and integrity

column	n_unique	missing	duplicate_non_null_values	unique_fraction
id	314168	0	0	1.0
member_id	0	314168	0	0.0

Exact duplicate rows detected: **0**.

10. LendingClub domain checks

These are heuristic checks. Exceptions require source documentation and contextual review; they are not necessarily errors.

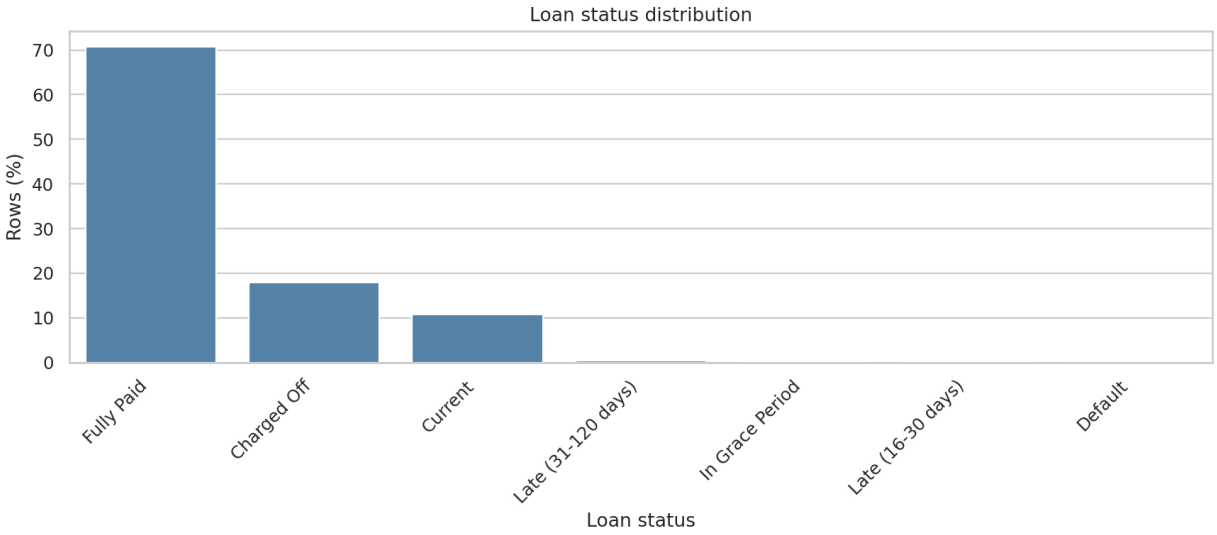
check	flagged_rows	flagged_pct	interpretation
funded_amnt > loan_amnt	0	0.0	Investigate; amount definitions or data vintage may explain exceptions.
funded_amnt_inv > funded_amnt	0	0.0	Investor-funded amount normally should not exceed total funded amount.
interest rate <= 0	0	0.0	Potential invalid values; check units and source.
interest rate > 100	0	0.0	Likely invalid or unit issue.
installment < 0	0	0.0	Negative installment requires investigation.
FICO low > FICO high	0	0.0	Range endpoints appear reversed.
FICO range width unusual	0	0.0	Review unusual FICO band widths.
DTI < 0	0	0.0	Negative debt-to-income values need review.
revol_util outside 0–200	0	0.0	Values above 100 may occur due to source conventions; inspect rather than auto-delete.
Missing loan_status	0	0.0	Missing outcome complicates supervised modeling.
Missing loan ID	0	0.0	Loan identifier missing.

11. Payment accounting consistency

check	flagged_rows	flagged_pct	note
total_rec_prncp > loan_amnt by > 1	0	0.0	May reflect adjustments, data definitions, or anomalies; investigate.
payment component mismatch > $2	39037	12.4255	Definitions/rounding may differ; compare source documentation.
investor outstanding principal > total outstanding principal by > $1	0	0.0	Check amount semantics and source version.

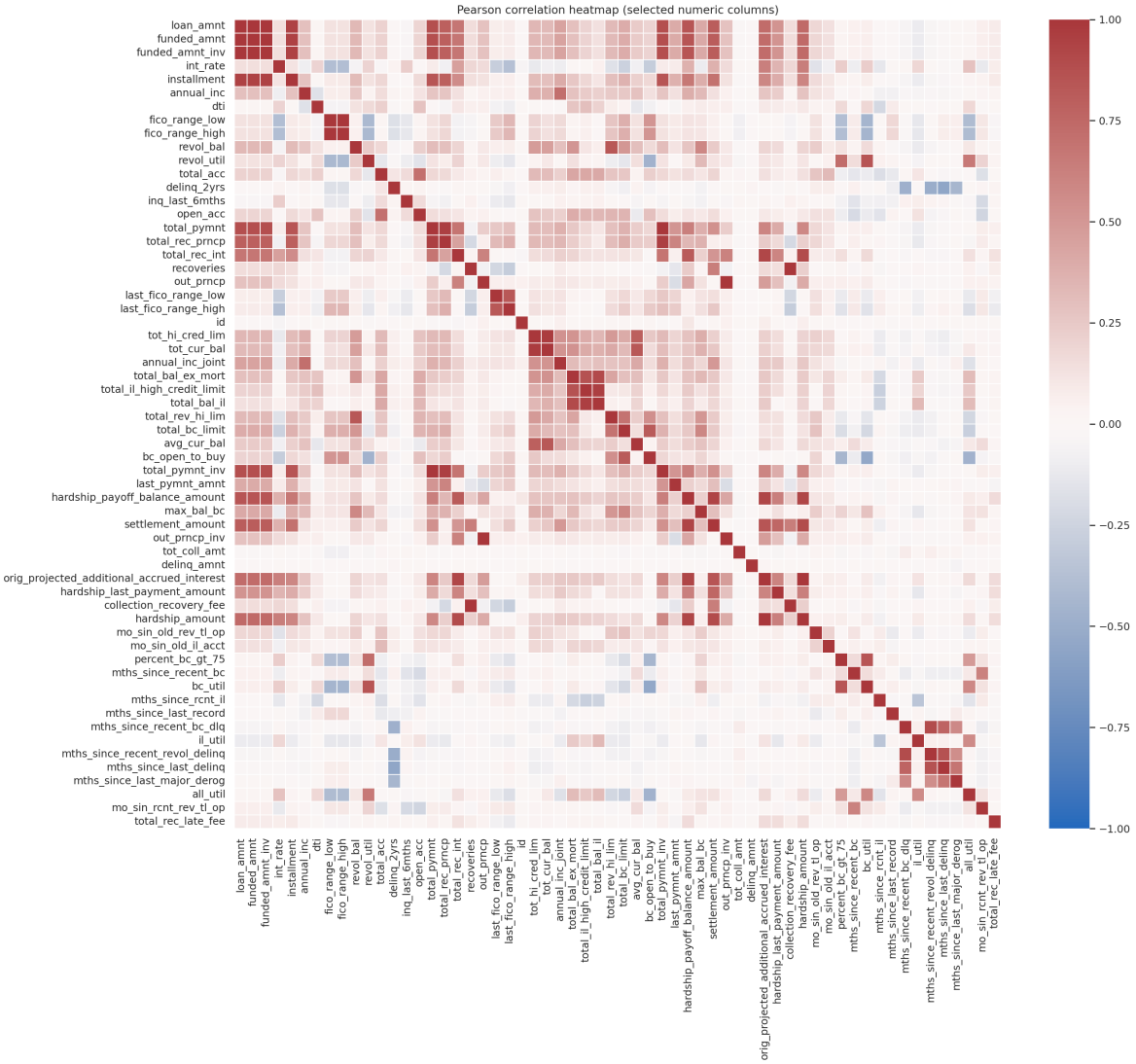
12. Loan status distribution

loan_status	count	percent
Fully Paid	222280	70.75195436836343
Charged Off	56159	17.87546790252349
Current	33891	10.787540424231622
Late (31-120 days)	1113	0.3542690535000382
In Grace Period	501	0.1594688192304754
Late (16-30 days)	223	0.0709811311145629
Default	1	0.0003183010363881745



13. Numeric correlation

Pearson correlation captures linear association and can be affected by outliers. Correlated variables may be redundant, but correlation alone does not prove that one feature should be removed.



feature_1	feature_2	correlation
loan_amnt	funded_amnt	1.0
orig_projected_additional_accrued_interest	hardship_amount	0.9999999999999988
fico_range_low	fico_range_high	0.9999999265750535
out_prncp	out_prncp_inv	0.9999981419243492
total_pymnt	total_pymnt_inv	0.9999947873043967
loan_amnt	funded_amnt_inv	0.9999937101707953
funded_amnt	funded_amnt_inv	0.9999937101707953
recoveries	collection_recovery_fee	0.9932284206759634
tot_hi_cred_lim	tot_cur_bal	0.9712946809102336
total_il_high_credit_limit	total_bal_il	0.9586317195288154
total_rec_prncp	total_pymnt_inv	0.9554732928127008
total_pymnt	total_rec_prncp	0.9553973507640832
funded_amnt	installment	0.9421050936003116
loan_amnt	installment	0.9421050936003116
funded_amnt_inv	installment	0.9420861563654166
hardship_payoff_balance_amount	settlement_amount	0.9360643511527672
hardship_payoff_balance_amount	hardship_amount	0.9266244951403876
hardship_payoff_balance_amount	orig_projected_additional_accrued_interest	0.9247131704089314
total_rec_int	orig_projected_additional_accrued_interest	0.9169687087330308
total_bal_ex_mort	total_bal_il	0.9042258312444521
total_rec_int	hardship_amount	0.8979661296351875

feature_1	feature_2	correlation
settlement_amount	hardship_amount	0.8920614456802901
mths_since_recent_bc_dlq	mths_since_recent_revol_delinq	0.8911395744954386
funded_amnt_inv	total_pymnt_inv	0.8792510263578162
funded_amnt_inv	total_pymnt	0.8792358653372058
funded_amnt	total_pymnt_inv	0.8792092573530242
loan_amnt	total_pymnt_inv	0.8792092573530242
funded_amnt	total_pymnt	0.879205008735876
loan_amnt	total_pymnt	0.879205008735876
total_bal_ex_mort	total_il_high_credit_limit	0.8647649938786017

14. Modeling-readiness and leakage audit

Review all flagged fields against the intended prediction time. For an origination-time risk model, payment history, recoveries, last payment, settlement, hardship, outstanding principal, and post-loan FICO are likely unavailable or leakage-prone.

column	audit_flag	reason
id	REVIEW	Identifier / link / granular geography
member_id	REVIEW	Identifier / link / granular geography
issue_d	REVIEW	Date field: define availability at prediction time
loan_status	REVIEW	Potential post-origination / outcome leakage
url	REVIEW	Identifier / link / granular geography
zip_code	REVIEW	Identifier / link / granular geography
earliest_cr_line	REVIEW	Date field: define availability at prediction time
out_prncp	REVIEW	Potential post-origination / outcome leakage
out_prncp_inv	REVIEW	Potential post-origination / outcome leakage
total_pymnt	REVIEW	Potential post-origination / outcome leakage
total_pymnt_inv	REVIEW	Potential post-origination / outcome leakage
total_rec_prncp	REVIEW	Potential post-origination / outcome leakage
total_rec_int	REVIEW	Potential post-origination / outcome leakage
total_rec_late_fee	REVIEW	Potential post-origination / outcome leakage
recoveries	REVIEW	Potential post-origination / outcome leakage
collection_recovery_fee	REVIEW	Potential post-origination / outcome leakage
last_pymnt_d	REVIEW	Potential post-origination / outcome leakage; Date field: define availability at prediction time
last_pymnt_amnt	REVIEW	Potential post-origination / outcome leakage
next_pymnt_d	REVIEW	Potential post-origination / outcome leakage; Date field: define availability at prediction time
last_credit_pull_d	REVIEW	Date field: define availability at prediction time
last_fico_range_high	REVIEW	Potential post-origination / outcome leakage
last_fico_range_low	REVIEW	Potential post-origination / outcome leakage
chargeoff_within_12_mths	REVIEW	Potential post-origination / outcome leakage
sec_app_fico_range_low	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_fico_range_high	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_earliest_cr_line	REVIEW	Secondary applicant field; often structurally missing for individual applications; Date field: define availability at prediction time
sec_app_inq_last_6mths	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_mort_acc	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_open_acc	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_revol_util	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_open_act_il	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_num_rev_accts	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_chargeoff_within_12_mths	REVIEW	Potential post-origination / outcome leakage; Secondary applicant field; often structurally missing for individual applications
sec_app_collections_12_mths_ex_med	REVIEW	Secondary applicant field; often structurally missing for individual applications
sec_app_mths_since_last_major_derog	REVIEW	Secondary applicant field; often structurally missing for individual applications
hardship_flag	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_type	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_reason	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_status	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_amount	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_start_date	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information; Date field: define availability at predicti
hardship_end_date	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information; Date field: define availability at predicti
payment_plan_start_date	REVIEW	Potential post-origination / outcome leakage; Date field: define availability at prediction time
hardship_length	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_dpd	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_loan_status	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_payoff_balance_amount	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_last_payment_amount	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
debt_settlement_flag	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
debt_settlement_flag_date	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information; Date field: define availability at predicti

column	audit_flag	reason
settlement_status	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
settlement_date	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information; Date field: define availability at predicti
settlement_amount	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
settlement_percentage	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
settlement_term	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
last_pymnt_d__parsed	REVIEW	Potential post-origination / outcome leakage
next_pymnt_d__parsed	REVIEW	Potential post-origination / outcome leakage
sec_app_earliest_cr_line__parsed	REVIEW	Secondary applicant field; often structurally missing for individual applications
hardship_start_date__parsed	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
hardship_end_date__parsed	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
payment_plan_start_date__parsed	REVIEW	Potential post-origination / outcome leakage
debt_settlement_flag_date__parsed	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information
settlement_date__parsed	REVIEW	Potential post-origination / outcome leakage; Special post-origination program / settlement information

15. Recommended next steps

1. Confirm the exact data vintage, source documentation, licensing, and intended analysis question.
2. Define the prediction time (e.g. loan origination) and outcome horizon before constructing a target.
3. Separate completed outcomes from active/censored loans; document how `Current` and late statuses are handled.
4. Exclude IDs, URLs, and direct identifiers from predictive features; treat ZIP and employer text carefully.
5. Build a data dictionary with units, allowed values, and time availability for each feature.
6. Investigate high-missingness columns and distinguish structural missingness from data defects.
7. Split train/test by time where appropriate; fit imputers, encoders, scalers, and feature selection only on training data.
8. Check class imbalance, calibration, PR-AUC/ROC-AUC, threshold-specific costs, and fairness/stability across groups.
9. Keep a reproducible preprocessing pipeline and log all exclusions and transformations.

Appendix: output files

This notebook writes CSV tables and PNG plots to the output folder. The PDF is a compact profile report; use the CSV exports for detailed review and follow-up analysis.